

Mechanistic insights on the synergistic roles of glass content and curing temperature in reaction kinetics and microstructural evolution of alkali-activated lunar regolith simulants

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ABSTRACT

The highly reactive glass phase in lunar regolith is a critical yet insufficiently understood component. Consequently, neglecting its influence compromises the fidelity of performance predictions for alkali-activated lunar regolith materials. This study revealed the synergistic mechanism of glass content and curing temperature on the reaction pathways, microstructural evolution, and mechanical properties of alkali-activated lunar regolith simulants. Multi-scale characterization revealed that curing at 60 °C promoted continuous structural development, evidenced by decreased diffraction peaks of regularly shaped zeolites, increased bridging oxygen content, and an upward shift in Si/Al binding energy. The coexistence of N-A-S-H gel with well-ordered crystals resulted in progressive pore refinement and steady strength gain. In contrast, at 90 °C, samples with low glass content developed disordered petal-shaped zeolites within the matrix, which introduced microcracks and caused long-term strength regression. Prolonged curing further promoted continuous zeolite crystallization, reduced bridging oxygen content, and led to gel-network depolymerization. Conversely, samples with high glass content effectively compromised harmful crystallization by forming a highly stable gel network, enabling continuous microstructural densification and strength improvement. These findings established a materials design principle for extraterrestrial construction, providing a scientific pathway toward durable and resource-efficient in-situ lunar building.

1. Introduction

As Earth's only natural satellite, the Moon serves as both a potential outpost for deep space exploration and a strategic gateway to the broader solar system [1,2]. The abundance of resources on the lunar surface has motivated global efforts toward establishing lunar bases and achieving human settlement [3–5]. However, constructing a permanent lunar base faces exceptional challenges, mainly due to its extreme environment. Without a substantial atmosphere, the Moon exists in an ultra-high vacuum and undergoes drastic cyclic thermal shocks, with temperatures ranging from approximately –180 °C to 120 °C [4,6–8]. The Moon is also continuously exposed to micrometeoroid impacts and intense radiation from solar wind and cosmic rays [9,10]. Additionally, the Moon's surface gravity, only about one-sixth of Earth's, poses unique

challenges to human health and engineering design [11–13]. Given the extremely high cost of transporting materials from Earth to the Moon, the adoption of ISRU technology has become an essential strategy for sustainable lunar construction. To address these challenges, various lunar regolith solidification methods have been proposed. Based on their hardening mechanisms, existing lunar regolith solidification methods could be classified into three main types: adhesive bonding (such as sulfur concrete and polymers) [14,15]; high-temperature sintering or melting [16–20]; and hydration-based (such as cementitious binders and geopolymers) [4,21–28]. Nevertheless, each approach has significant limitations. For example, sulfur concrete is promising for lunar use because it requires no water or Earth-sourced binders. Yet, it was constrained by significant brittleness and progressive degradation under thermal cycling, restricting its use to temperatures below 120 °C [14].

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Portland cement-based concrete demonstrated high mechanical strength, excellent low-temperature resistance, and maintained structural integrity in low gravity. But most of its raw materials must be transported from Earth, resulting in high costs. Polymer bonding depends mainly on mechanical interlocking with lunar regolith particles and tends to debond at the interface under extreme lunar conditions. Moreover, its dependence on Earth-supplied raw materials constrained its suitability for true in-situ applications. High-temperature sintering or melting requires substantial energy consumption, complex equipment, and may fail under extreme conditions, rendering them unsuitable for large-scale use. In addition to these chemical and thermal methods, mechanical approaches have also been explored for lunar regolith solidification. For instance, Geng et al. proposed a novel 3D printing scheme based on powder extrusion and passive roll-pressing, achieving flexural strengths of 2–5 MPa with extremely low binder utilization through mechanical compaction [29]. In contrast, lunar regolith-based geopolymers offer significant advantages for extraterrestrial construction, exhibiting key properties for the lunar environment such as high mechanical strength, excellent durability, and stability under extreme temperature fluctuations and radiation [30,31]. Critically, their synthesis aligns closely with ISRU principles. The required alkali activators could potentially be extracted from lunar minerals, the regolith itself serves as an abundant aluminosilicate precursor, and the necessary water could be obtained from polar ice deposits. In recent years, alkali-activated lunar regolith materials have garnered significant attention in the field of in-situ construction on the Moon. Geng et al. developed a silicate-lunar regolith composite that self-hardens in vacuum through dehydration of silica clusters, achieving 29 MPa with only 4 wt% binder and no heating, a mechanism fundamentally different from alkali-activated geopolymerization, which relied on the dissolution of aluminosilicates and formation of new reaction products under sealed thermal curing [32]. Turning to alkali-activated systems specifically, Wu et al. systematically investigated various lunar regolith-activator combinations through integrated experimental and statistical analysis. Their findings established key principles for lunar geopolymer synthesis: sodium hydroxide was optimal for activating glass-rich regolith, whereas sodium silicate was more suitable for glass-free regolith. They also determined that thermal curing above 60–80 °C was essential to achieve applicable 24-h strength [33]. Complementing these formulation guidelines, Geng et al. elucidated the underlying solidification mechanisms under different environmental conditions. They revealed that under sealed curing, solidification originated from geopolymerization driven by strong alkalis and elevated temperatures (>64–75 °C), whereas under vacuum exposure, the dehydration of high-modulus sodium silicate became the dominant solidification pathway [34]. Together, these studies provided practical guidance for selecting suitable activator-regolith combinations and offered mechanistic insights into how environmental conditions influence the solidification process.

Beyond the predominant crystalline phases, lunar regolith contains a substantial fraction of glassy phases, formed by the Moon's extreme surface environment. These amorphous phases are primarily generated from meteoritic impacts and volcanic eruptions. Impacts cause localized melting of minerals, and the resulting amorphous melt is rapidly quenched in the lunar vacuum to form diverse glassy morphologies. Similarly, volcanic eruptions produce fine vitrified droplets through rapid cooling of ejected magma. A distinct form, agglutinitic glass, forms as impact melts bind surrounding mineral and rock fragments. Notably, the abundance of such glasses varies markedly across the lunar surface. For example, Chang'E-5 samples contain between 8.3% and 20.0% glassy phase, whereas Apollo regolith records substantially higher abundances, ranging from 25.4% to 72.3% [35–43].

Structurally, lunar glasses are amorphous, exhibiting short-range order but long-range disorder as a result of rapid quenching from a melt. This process results in an interconnected network of Si-O and Al-O tetrahedra that contains numerous broken and unsaturated bonds. The high-energy state of this network drives its depolymerization in an

alkaline environment, releasing highly reactive silicate and aluminate species. These subsequently undergo polycondensation, ultimately forming a continuous three-dimensional gel matrix.

Despite the widely acknowledged significance of the glass phase in lunar regolith, the synergistic interplay between glass content and curing temperature, specifically how they jointly govern reaction kinetics, phase evolution, and material performance, remains largely unexplored. This mechanistic gap has significant practical implications: without understanding how glass content and temperature interact, it is impossible to optimize curing conditions for regolith with varying compositions, directly hindering reliable lunar construction. From a scientific perspective, this knowledge gap also limits the ability to predict and understand geopolymerization behaviors in lunar systems, where the wide variation in glass content across different landing sites introduces additional complexity. To bridge this critical knowledge gap, this study systematically investigated the effects of varying glass-phase content on the properties of NaOH-activated lunar regolith simulant geopolymers under different thermal curing temperatures. Through comprehensive phase and microstructural analyses, this study elucidated the critical relationships between compositional variables, reaction mechanisms, and the macroscopic properties of these extraterrestrial cementitious materials.

2. Methods and materials

2.1. The preparation of basalt glass

Basalt powder was melted in a lifting furnace under ambient air conditions. The temperature was first raised to 1200 °C at 10 °C min⁻¹, and then further increased to 1400 °C at 8 °C min⁻¹. The melt was maintained at 1400 °C for 1 h to ensure complete melting and homogenization. Subsequently, it was rapidly quenched in water to form amorphous basalt glass. The obtained glass was ground using a vibratory mill and sieved to collect particles smaller than 75 µm for later use.

2.2. Geopolymer preparation

The geopolymers were prepared as follows. First, basalt powder and basalt glass (both below 75 µm) were dry-mixed according to the proportions shown in Table 1 to form a homogeneous solid precursor. An 8 mol/L alkaline activator was prepared by dissolving sodium hydroxide pellets (≥96% purity) in deionized water. The mass ratio of the alkaline activator to the solid mixture (liquid-to-solid ratio) was set at 0.28. The activator was then poured into the solid mixture and blended in two stages: first at 150 ± 5 rpm for 2 min, then at 300 ± 10 rpm for 2 min. The resulting paste was cast into 20 mm × 20 mm × 20 mm molds and vibrated for 60 s to remove air bubbles. To prevent moisture loss during curing, each mold was immediately wrapped with multiple layers of polyethylene film and tightly secured. Samples for different curing ages (1, 3, 7, and 28 days) were packaged separately in independent sealed containers, ensuring that the sealing integrity of remaining samples was not compromised when retrieving specimens at specific time points. Throughout the curing period, the plastic film remained visibly bulged due to the accumulation of internal vapor pressure, indicating that a

Table 1
The mix proportions of samples.

Sample	NaOH (g)	Water (g)	Basalt (g)	Glass (g)	Temperature (°C)
SH-1-60	8.15	25.45	120	0	60
SH-2-60	8.15	25.45	102	18	60
SH-3-60	8.15	25.45	84	36	60
SH-4-60	8.15	25.45	48	72	60
SH-1-90	8.15	25.45	120	0	90
SH-2-90	8.15	25.45	102	18	90
SH-3-90	8.15	25.45	84	36	90
SH-4-90	8.15	25.45	48	72	90

saturated vapor environment was sustained inside; no loosening or collapse of the film was observed before sample retrieval. At each designated curing age, only the corresponding samples were opened for testing, while all others remained undisturbed under sealed conditions until their respective testing ages. These procedures ensured that moisture evaporation was negligible during the entire curing period, and that the reaction pathway for each sample was preserved until the moment of testing. Finally, specimens were cured at different temperatures (60 °C and 90 °C) under ambient atmosphere to evaluate the influence of temperature and glass phase content on the reaction kinetics and mechanical properties of the geopolymers.

2.3. Characterization methods

The compressive strength was measured on cube specimens (20 mm × 20 mm × 20 mm) at curing ages of 1, 3, 7, and 28 days using a universal testing machine (TYE-3000, Wuxi Jianyi Experimental Instrument) at a loading rate of 0.5 mm/min. For each curing condition, two replicate specimens were prepared and tested. The reported compressive strength values represent the mean of the two replicates, with error bars in the figures indicating the standard deviation. After the compressive test, to more accurately and comprehensively reflect the overall microstructure of the sample, the central portions of the samples were taken, soaked in isopropyl alcohol to terminate hydration, and then dried in a vacuum oven. After drying, they were ground for subsequent testing.

A Quanta 450FEG type field emission scanning electron microscope (SEM) manufactured by FEI was used to characterize the morphology of the sample in high vacuum mode, which was coated with a layer of platinum, with an accelerating voltage of 15 kV and a working distance of 10 mm.

X-Ray Diffraction (XRD) was performed to identify the mineralogical phases of the samples. The phase composition of the samples was analyzed by X-ray diffractometer (D8-Advance, Bruker, Germany), with a test voltage of 35 kV and a test current of 30 mA at a scanning rate of 2°/min and a scanning range of 5–70°. Before XRD analysis, 10 wt% ZnO was used as an internal standard. The sample and ZnO were wet-ground in alcohol to below 75 µm and then dried in a vacuum oven at 40 °C for 12 h.

The Thermogravimetry-Derivative Thermogravimetry (TG-DTG) curves for samples were collected by a thermogravimetric analyzer (STA449F3, Netzsch, Germany) at a temperature rise rate of 10 °C/min, ranging from room temperature to 1000 °C. Nitrogen was used as a protective gas with a gas flow rate of 20 mL/min.

The chemical oxide composition of the samples was determined by X-ray fluorescence spectrometry using a ZETIUM spectrometer (Malvern Panalytical). The instrument is equipped with a 4 kW rhodium (Rh) end-window X-ray tube. Sample preparation followed a standard fusion procedure to mitigate particle size and mineralogical effects. Calibration was performed using certified reference materials, and matrix effects were corrected employing an empirical coefficient method, resulting in relative errors of less than 0.05% for major element analyses.

Fourier-transform infrared (FT-IR) spectra were collected using a Nicolet 6700 spectrometer. Measurements were performed in transmission mode within the mid-infrared spectral range of 4000–400 cm^{−1}. Data were collected at a spectral resolution of 0.019 cm^{−1} with a rapid scanning rate of 1 scan per second.

The pore structure characteristics of the geopolymer samples were analyzed using mercury intrusion porosimetry (MIP) with an AutoPore V 9505 instrument (Micromeritics Instrument Corporation, USA).

X-ray photoelectron spectroscopy (XPS) analysis was performed using a Thermo ESCALAB 250Xi system under ultra-high vacuum conditions. The energy scanning range of the spectrometer was 0–5000 eV, with a pass energy adjustable from 1 to 400 eV. The analysis spot size was varied between 20 and 900 µm as required. High-resolution spectra were collected for the C 1s, Si 2p, Al 2p, and O 1s core levels. The energy

resolution of the system was 0.45 eV, and the sensitivity reached 1.0 Mcps at 0.6 eV with a 650 µm spot size. Chemical imaging was conducted with a spatial resolution of 3 µm. All binding energies were referenced to the C 1s peak at 284.8 eV, and data processing was performed using the Advantage software suite.

3. Results

3.1. The characterization of the raw materials

The characterization of the raw basalt and its derived glass confirmed their suitability as geopolymer precursors. The basaltic powder used in this study was sourced from Nanjing, Jiangsu Province, China. Particle size analysis (Fig. 1a) revealed a continuous unimodal distribution, with D₁₀, D₅₀, and D₉₀ values of 1.355 µm, 9.663 µm, and 39.217 µm, respectively. This fine particle size distribution provided a high specific surface area, favoring rapid dissolution in alkaline activators and promoting geopolymerization kinetics. Quantitative XRD analysis (Fig. 1b) determined that the basalt was predominantly plagioclase (60.3 wt%) and augite (28.7 wt%), with subordinate forsterite (9.6 wt%) and minor amounts of ilmenite (0.3 wt%) and zeolite (0.9 wt%). Its low amorphous content (0.2 wt%) confirmed a highly crystalline nature. Notably, this mineral assemblage aligned closely with the dominant phases reported for genuine lunar mare regolith, particularly the Apollo mission basalts, validating its relevance as a mineralogical and chemical analog.

The XRD pattern of the melt-quenched product (Fig. 1e) confirmed complete amorphization, showing only a broad halo characteristic of a glassy structure. The complete absence of sharp crystalline diffraction peaks verified the transition to a long-range disordered, short-range ordered atomic arrangement, representing the definitive structural signature of a glassy state. This phase transformation was further supported by chemical analysis (Fig. 1c). Compared to the precursor basalt, the resultant glass exhibited a distinct signature marked by significant enrichment in FeO (12.904 wt%) and MgO (8.833 wt%) alongside relative depletion in Al₂O₃ (14.757 wt%).

3.2. Compressive strength

As illustrated in Fig. 2, under 60 °C curing conditions, the incorporation of the glassy phase significantly enhanced the compressive strength development, particularly at early ages. Strength at 1, 3, and 7 days rose monotonically with glass content up to 60%. The 60% glass sample (SH-4-60) reached a 1-day strength over ten times greater than the glass-free control. While the 28-day strength increased substantially from 11.9 MPa (SH-1-60) to 41.5 MPa (SH-3-60), the sample with the highest glass content (SH-4-60, 60%) showed only a marginal improvement to 42.6 MPa. Notably, the 60% glass sample (SH-4-60) exhibited an exceptional strength gain between 3 and 7 days, reaching 32.2 MPa, constituting most of its final 28-day strength.

When the curing temperature was elevated to 90 °C, the reaction kinetics accelerated further, resulting in nearly complete strength development within 7 days for all mixtures. The sample with 60% glass content (SH-4-90) performed best, attaining the highest 28-day strength of 49.4 MPa. In contrast, samples with lower glass content showed strength regression from 7 to 28 days. The comparative analysis between 60 °C and 90 °C curing revealed a critical interaction: although elevated temperatures generally accelerated the reaction, samples with lower glass content showed a slight decrease in strength at later ages, unlike those with high glass content.

3.3. X-ray diffraction

Fig. 3 presented the XRD patterns of samples cured at 60 °C (Figs. 3a) and 90 °C (Fig. 3b) for 7 and 28 days, together with the quantitative phase composition of representative samples with 30% glass phase

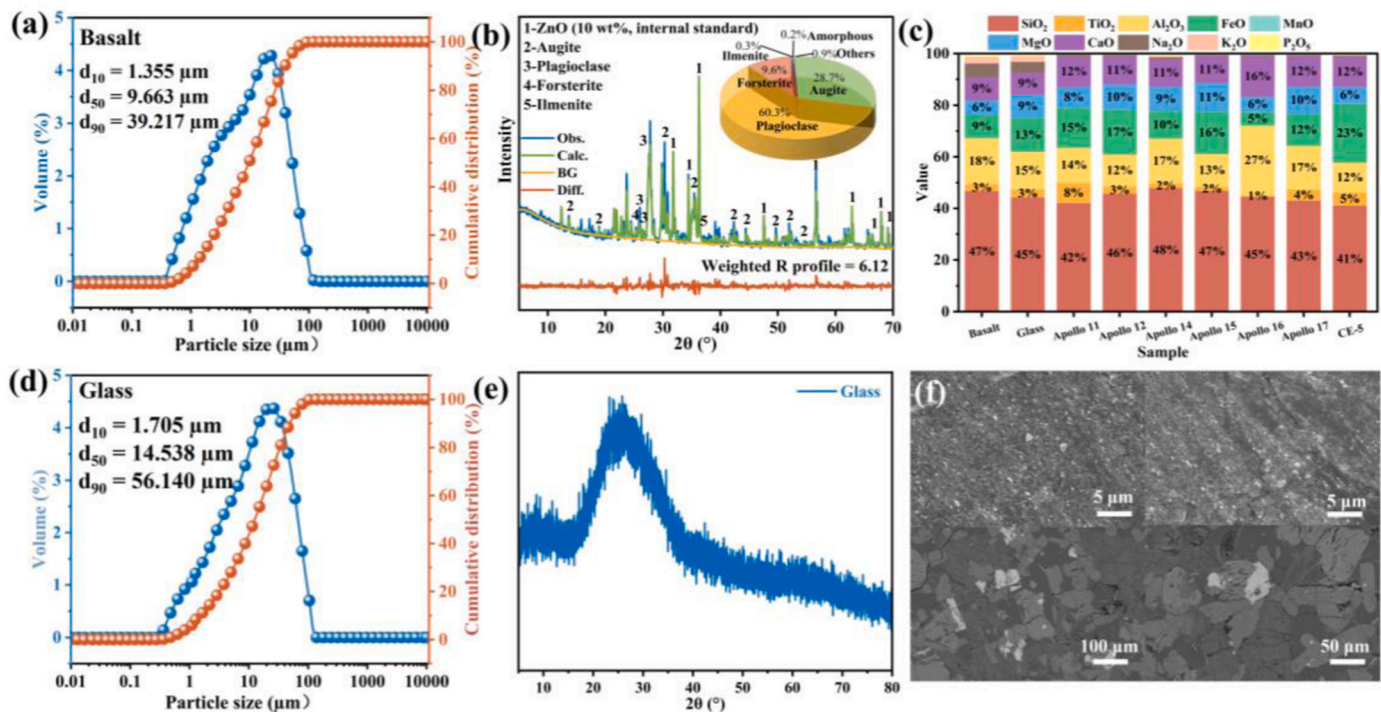


Fig. 1. Characterization of the raw basalt and prepared basalt glass; (a) Particle size distribution of the raw basalt material, (b) X-ray diffraction (XRD) pattern and quantitative analysis of the raw basalt, (c) Chemical oxide composition comparison of real lunar regolith, the raw basalt, and the synthesized basalt glass, (d) Particle size distribution of the basalt glass, (e) XRD pattern of the amorphous basalt glass, (f) SEM-BSE image of the raw basalt, illustrating its microstructure.

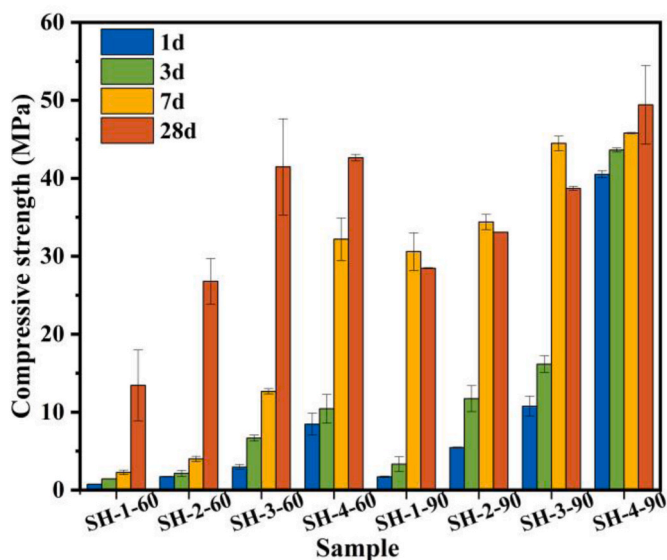


Fig. 2. Compressive strength development under different curing temperatures and glass contents.

content determined by Rietveld refinement with 10 wt% ZnO as an internal standard (Fig. 3c–f). At 60 °C (Fig. 3a), the XRD patterns of all samples were mainly composed of the crystalline phases of the original basalt precursor, primarily plagioclase, augite, and forsterite. Simultaneously, a distinct amorphous hump in the 2θ range of 20° – 40° was also observed, corresponding to N-A-S-H gel formation and confirming the activation of the glass phase in the alkaline environment. The intensity of this amorphous signal increased with glass content, along with higher background scattering, indicative of enhanced gel formation. Several zeolite phases were also detected. While the glass-free control sample

(SH-1) contained zeolite ZK-14 as the main reaction product, the introduction of glass shifted the zeolite crystallization toward Na-P1. Notably, the diffraction peaks of zeolites weakened from 7 to 28 days, suggesting that these zeolites formed at 60 °C were not fully stable. Quantitative analysis of the 30% glass samples (Fig. 3c–d) confirmed this observation: total zeolite content (Na-P1 and analcime) decreased from 10.9% at 7 days to 9.0% at 28 days, while amorphous content increased from 36.6% to 38.6%, indicating that the dissolved zeolites were reincorporated into the growing gel network.

In contrast, under 90 °C curing (Fig. 3b), ZK-14 zeolite reappeared in the control sample. With 15% glass content, zeolite ZK-14 coexisted with a minor amount of Na-P1 zeolite. When the glass content reached 30%, characteristic peaks of analcime appeared. Unlike the behavior observed at 60 °C, the intensity of these zeolite peaks remained stable over time, rather than decreasing. Correspondingly, quantitative analysis of the 30% glass samples at 90 °C (Fig. 3e–f) showed that total zeolite content remained stable at 8.3% at both 7 and 28 days, while amorphous content increased from 43.1% to 44.3%, demonstrating that zeolites were stable at elevated temperatures and gel formation proceeds independently. This temperature-dependent behavior confirmed that zeolite stability was governed by curing temperature: metastable at 60 °C, stable at 90 °C (see Table 2).

These observations were consistent with previous studies reporting that a curing temperature of 90 °C was more effective for the formation of Na-P1 zeolite in geopolymer systems than 60 °C, with zeolite diffraction peaks intensifying progressively as curing time extended [44]. It had also been documented that the crystallization of zeolites could lead to a partial reduction in compressive strength [44]. The XRD and quantitative analyses revealed a clear temperature-dependent phase evolution: at 60 °C, zeolites dissolved over time and contributed to gel formation, whereas at 90 °C, they remained stable throughout curing.

3.4. Thermogravimetry-derivative thermogravimetry

Fig. 4 presented the TG-DTG curves of samples cured at 60 °C and

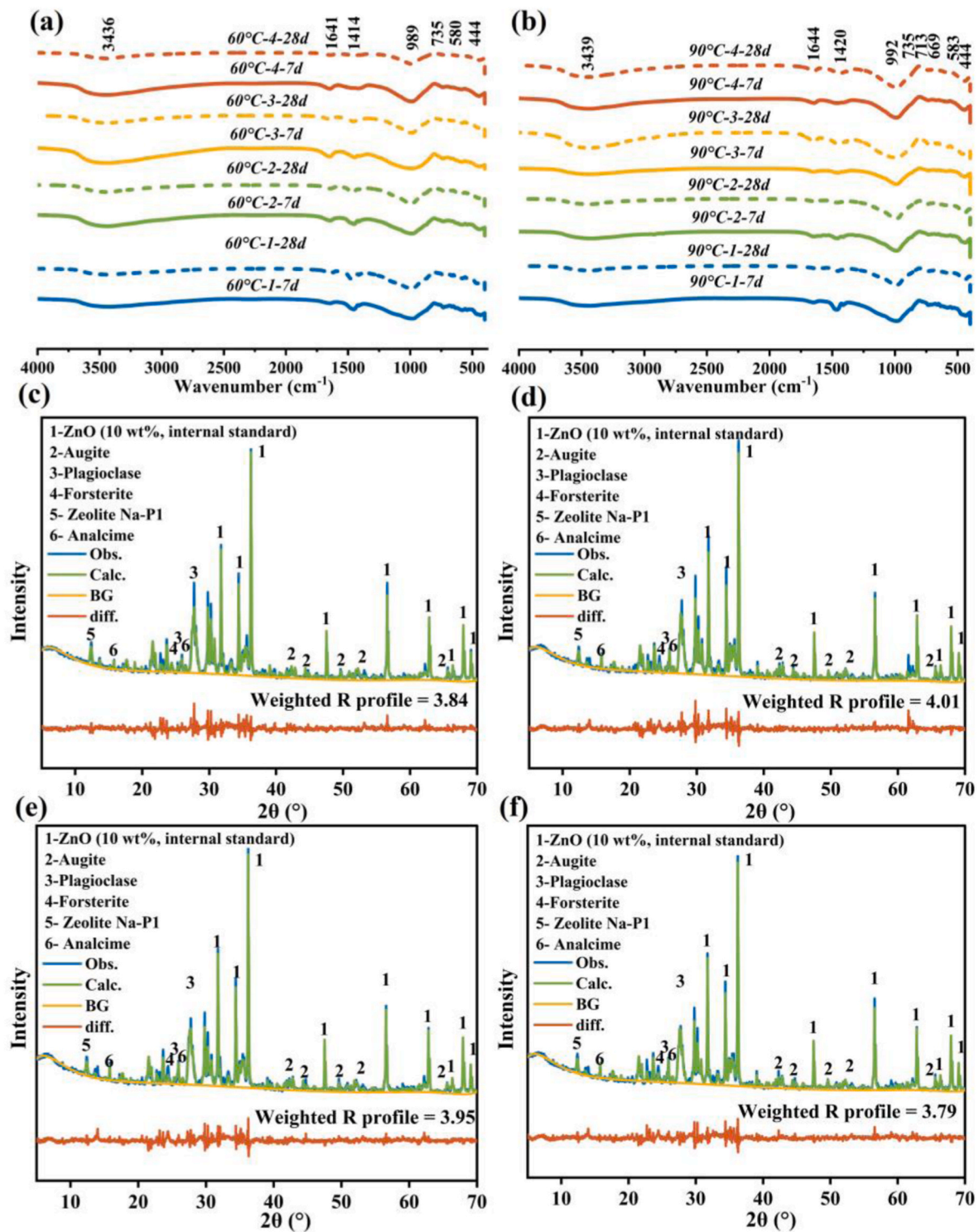


Fig. 3. The XRD patterns of the samples cured at (a) 60 °C for 7 and 28 days and (b) 90 °C for 7 and 28 days. (c-f) Quantitative phase composition (Rietveld refinement with 10 wt% ZnO internal standard) of samples with 30% glass phase content cured at: (c) 60 °C-7d; (d) 60 °C-28d; (e) 90 °C-7d; (f) 90 °C-28d.

Table 2

Quantitative phase composition of samples with 30% glass phase content cured at different temperatures determined by Rietveld refinement.

Sample	Amorphous	Zeolite Na-P1	Analcime	Others	ZnO
60 °C-3-7d	36.6%	9.9%	1%	42.5%	10%
60 °C-3-28d	38.6%	7.4%	1.6%	42.4%	10%
90 °C-3-7d	43.1%	6.7%	1.6%	38.6%	10%
90 °C-3-28d	44.3%	6.6%	1.7%	37.4%	10%

90 °C for 3, 7, and 28 days. The mass loss peak observed around 100 °C could be attributed to the removal of physically adsorbed water from both the zeolitic phases and the N-A-S-H gel. Previous studies have shown that in sample rich in N-A-S-H gel, the removal of free water occurred at relatively higher temperatures, reflecting the strong water-holding capacity of the gel structure [45]. As shown in Fig. 4, the mass loss above 100 °C increased progressively with glass phase content across all curing ages (3, 7, and 28 days). This trend indicated a greater

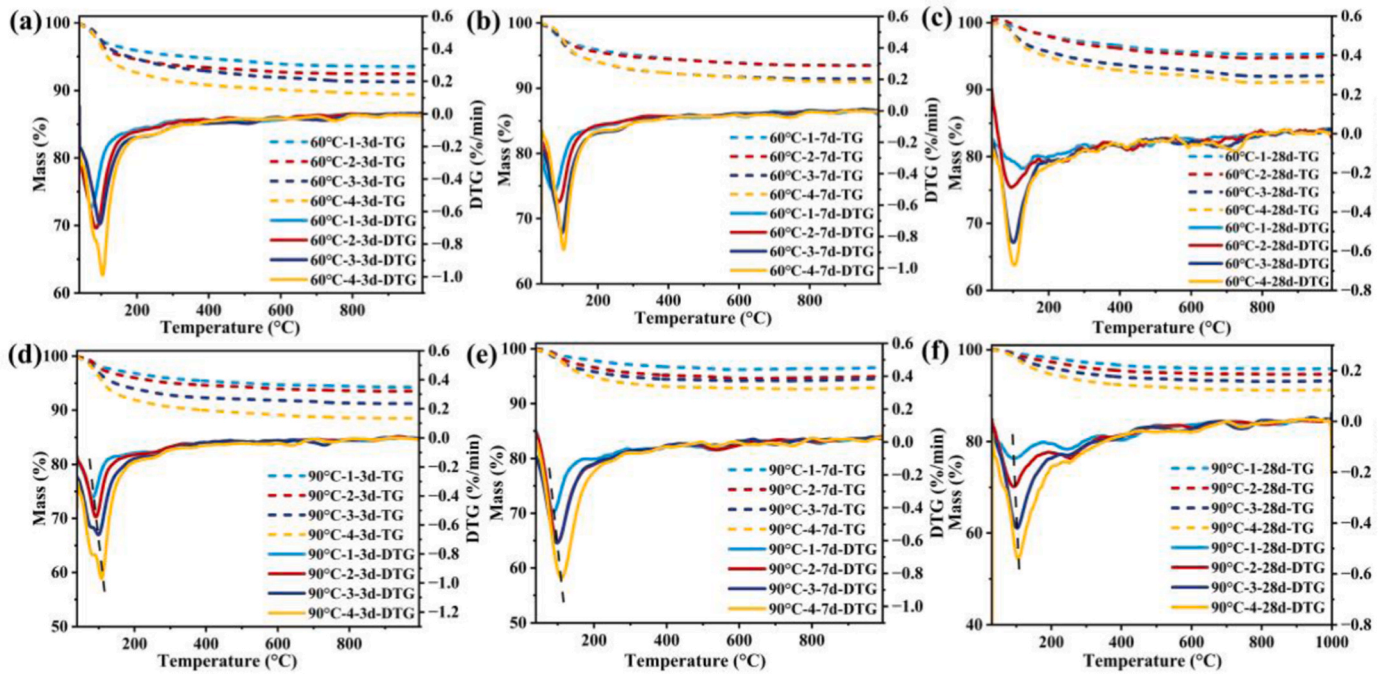


Fig. 4. TG-DTG curves of the samples cured at 60 °C for (a) 3, (b) 7, and (c) 28 days and cured at 90 °C for (d) 3, (e) 7, and (f) 28 days.

amount of N-A-S-H gel formation in high-glass-content samples. Concurrently, the corresponding DTG peak shifted toward higher temperatures, further corroborating that the structural development and maturation of the gel phase enhance the binding of water molecules within the matrix. In the temperature range of 200–600 °C, a broad characteristic peak appeared, corresponding to the removal of more strongly bound structural water from the N-A-S-H gel and the zeolitic phases. The dehydroxylation of these structurally incorporated -OH groups required higher energy, consistent with their stronger bonding within the aluminosilicate network.

3.5. Mercury intrusion porosimetry

Fig. 5a and c presented the cumulative pore volume curves of samples cured at 60 °C for 7 and 28 days. With increasing glass-phase content, the curves shifted progressively downward, reflecting a significant reduction in total porosity. This reduction was attributed to the enhanced reactivity of the glass phase, which increased the formation of reaction products that continuously filled pores and densified the microstructure. Pores were classified as macropores (>1000 nm), capillary pores (100–1000 nm), mesopores (10–100 nm), and gel pores (<10 nm). Among these, only macropores and capillary pores were considered to have a substantial negative effect on compressive strength.

Notably, although the total porosity of SH-3-60 remained nearly unchanged and that of SH-4-60 even increased slightly from 7 to 28 days (Fig. 5h), both samples exhibited compressive strength gain over the same period. Pore structure analysis resolved this apparent contradiction (Fig. 5b and d). In SH-3-60, the combined volume fraction of pores (macropores and capillary pores) decreased from 76% to 70%, while in the 60% glass sample (SH-4-60), it declined more markedly from 75% to 63%. These indicated that the observed increase in total porosity was largely due to the growth of gel pores and mesopores, which were associated with continued hydration and do not compromise mechanical performance. Hence, the underlying microstructural evolution supported, rather than hindered, sustained strength development.

The mercury intrusion cumulative curves (Fig. 5e and g) corroborated these trends. All curves exhibited a characteristic steep rise at low pressures, corresponding to rapid filling of fine pores, followed by a

plateau at high pressures that signaled the saturation of accessible pore volume. At each curing stage, the curve shifted downward systematically with increasing glass phase content, providing direct visual evidence of progressive porosity reduction.

The corresponding logarithmic differential intrusion curves (Fig. 5f) revealed distinct pore size distribution pathways. The glass-free reference showed a sharp unimodal peak that shifted from approximately 550 nm at 7 days to about 430 nm at 28 days, reflecting progressive pore refinement. With 15% glass (SH-2-60), a relatively narrow peak near 430 nm at 7 days broadened significantly by 28 days and shifted to around 350 nm. With 30% glass (SH-3-60), the distribution was already broad at 7 days (peak near 550 nm) and evolved into a nearly flat distribution by 28 days. While the 60% glass sample (SH-4-60) exhibited a notably flat differential curve, displaying only a faint, broad feature near 360 nm at 7 days, which further widened and flattened by 28 days, reflecting evolution toward a highly uniform, dense microstructure.

At 90 °C, a similar overall trend of porosity reduction with higher glass content was observed in the cumulative curves (Fig. 6a–c, e, g). However, pore size distributions (Fig. 6f) revealed critical differences. The glass-free sample (SH-1-90) maintained a dominant peak around 550 nm, with intensity increasing slightly over time, reflecting unstable pore coarsening linked to rapid zeolite crystallization that competed with gel formation and introduced defects. This further explained its strength regression. Samples with 15% and 30% glass (SH-2-90 and SH-3-90) exhibited a decrease in the main peak intensity along with broadening toward larger pores with extending curing, signaling persistent microstructural instability and rising porosity. In contrast, the sample containing 60% glass phase (SH-4-90) developed a bimodal pore structure at 7 days, with a prominent peak near 200 nm and a secondary peak around 20 nm, revealing an early network dominated by fine pores. As curing proceeded, the 200 nm peak diminished and broadened without significant coarsening, signaling a transition toward a more homogeneous and denser microstructure. This evolution, coupled with reduced porosity and enhanced compressive strength, demonstrated that sufficient glass phase secured the dominance of gel-forming reactions, which filled pores and sustained densification, ultimately ensuring superior long-term performance.

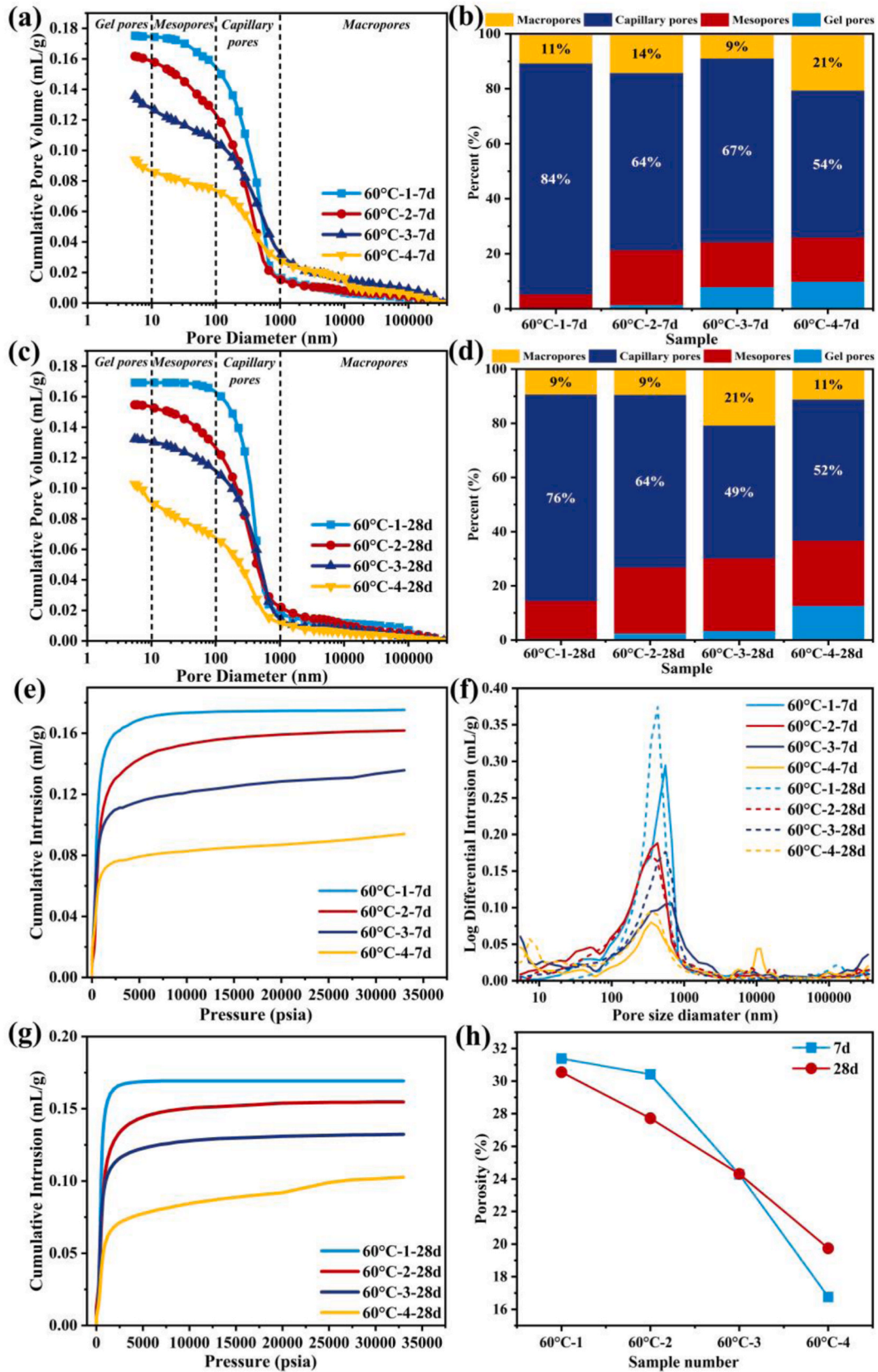


Fig. 5. Pore structure characterization by MIP; (a), (c) Cumulative pore volume distribution (CPVD) results for samples cured at 60 °C for 7 and 28 days, respectively, (b), (d) Pore volume results of samples cured at 60 °C for 7 and 28 days respectively, (e), (g) Cumulative intrusion volume under applied pressure for samples cured at 60 °C for 7 and 28 days, respectively, (f) Log differential intrusion volume versus pore diameter, (h) Comparison of total porosity between the 7- and 28-day curing periods.

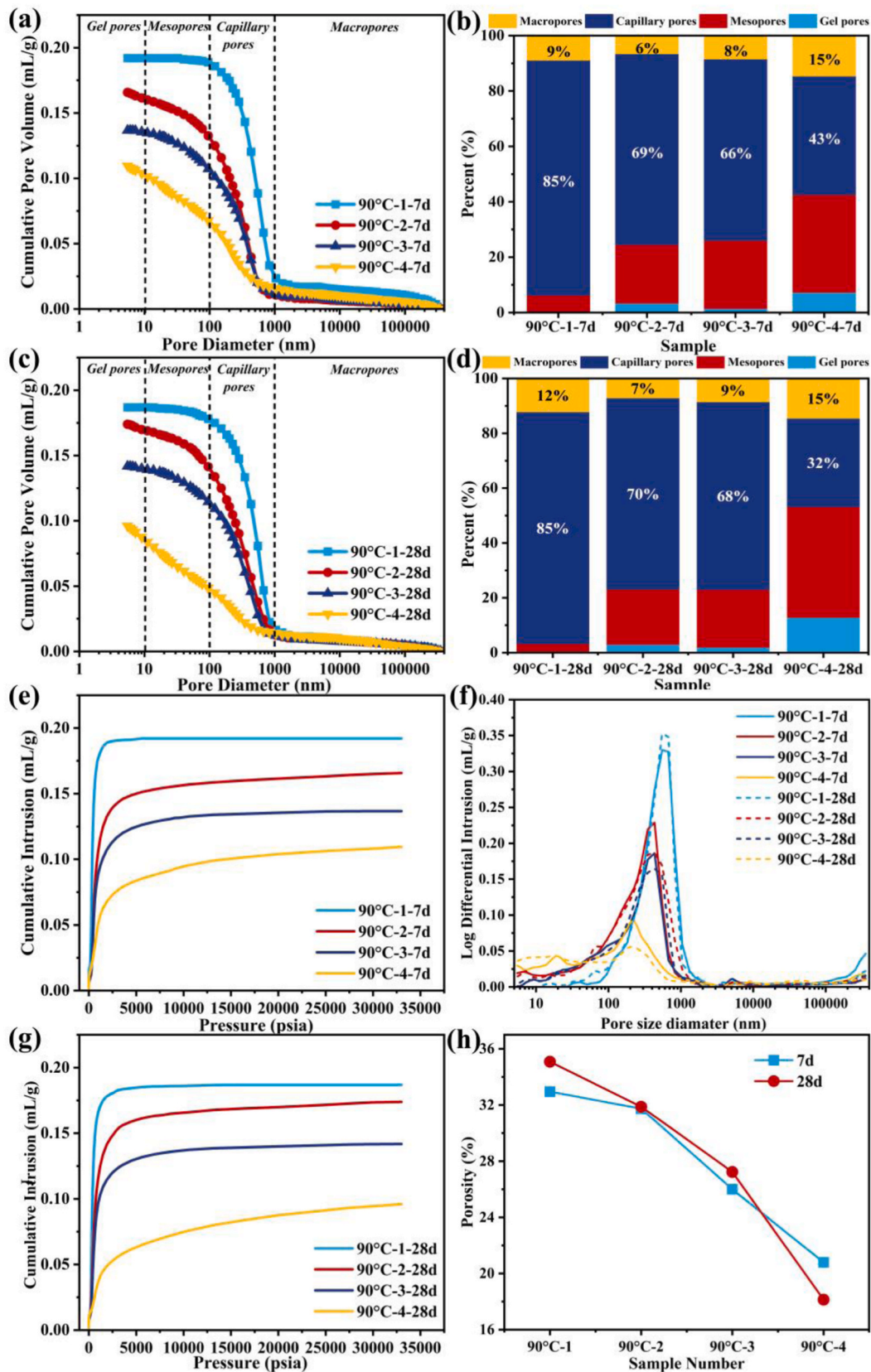


Fig. 6. Pore structure characterization by MIP; (a), (c) Cumulative pore volume distribution (CPVD) results for samples cured at 90 °C for 7 and 28 days, respectively, (b), (d) Pore volume results of samples cured at 90 °C for 7 and 28 days respectively, (e), (g) Cumulative intrusion volume under applied pressure for samples cured at 90 °C for 7 and 28 days, respectively, (f) Log differential intrusion volume versus pore diameter, (h) Comparison of total porosity between the 7- and 28-day curing periods.

3.6. X-ray photoelectron spectroscopy

XPS analysis was employed to investigate the chemical evolution of samples under varying conditions of glass phase content and curing temperature (Fig. 7). The C 1s peak at 284.8 eV was used for charge correction. Deconvolution of the O 1s spectra enabled quantification of three distinct oxygen environments: bridging oxygen (BO, Si/Al-O-Si/Al), non-bridging oxygen (NBO, Si/Al-O-Ca²⁺/Na⁺), and silanol

groups (Si-OH). The following analyses detailed the progressive variations in Si 2p, Al 2p, and O 1s spectra across different samples.

For the glass-free sample cured at 90 °C (Fig. 7a and b), the Si 2p_{3/2} binding energy decreased significantly from 102.1 eV at 7 days to 98.75 eV at 28 days with a concurrent slight decrease in Al 2p_{3/2} (74.02 eV to 73.97 eV). This downward shift indicated increased electron density around Si and Al nuclei, signaling depolymerization of the aluminosilicate framework. The deconvolution of O 1s spectra revealed a marked

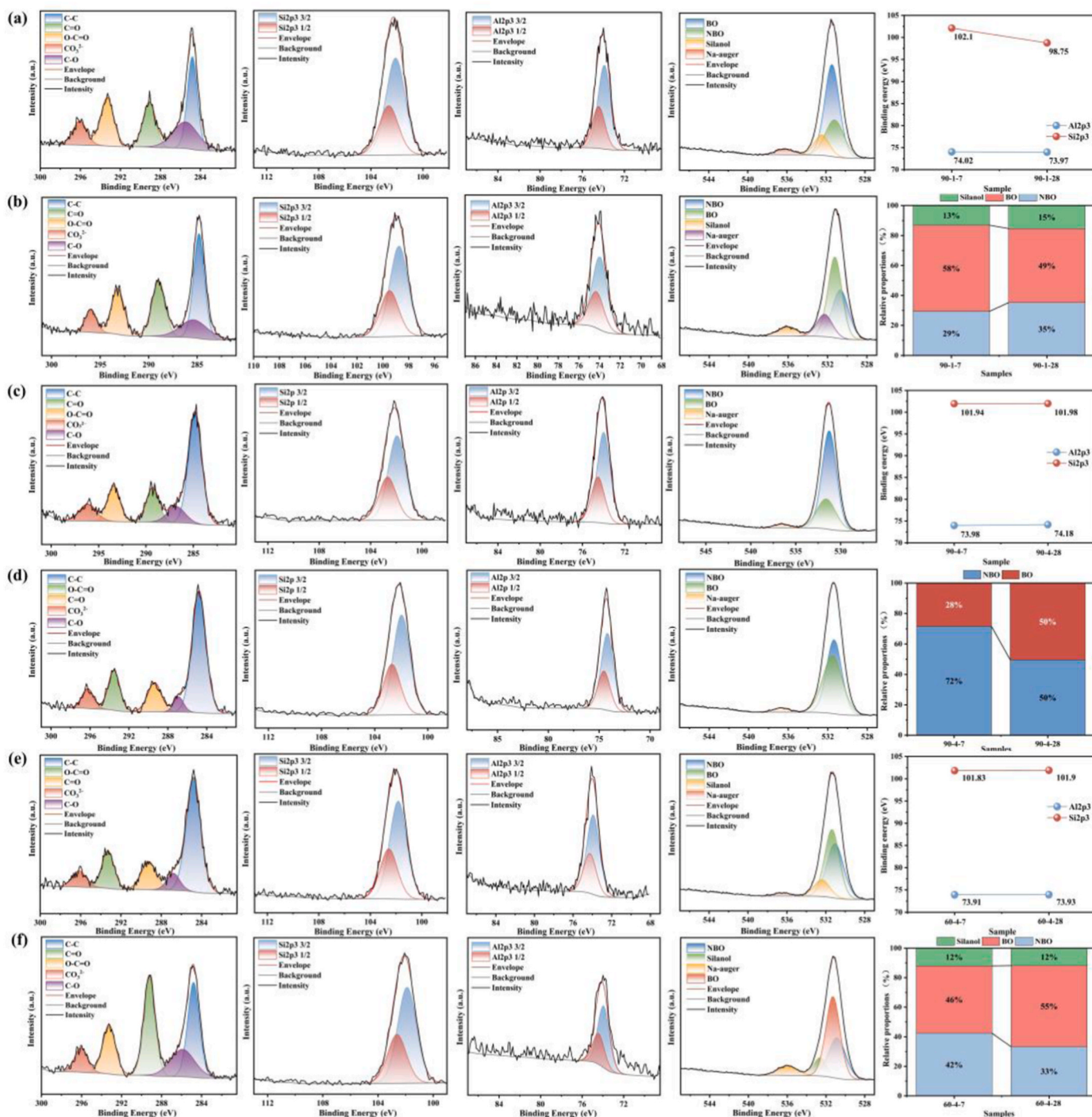


Fig. 7. XPS spectral deconvolution and fitting analysis of samples under varying compositions and curing conditions. High-resolution spectra correspond to C 1s, Si 2p, Al 2p, and O 1s regions, illustrating the evolution of chemical bonding environments as a function of curing time and temperature; XPS spectra and corresponding peak-fitting results of the glass-phase-free sample cured at 90 °C for (a) 7 days and (b) 28 days, XPS spectra and peak deconvolution of the sample containing 60% glass phase cured at 90 °C for (c) 7 days and (d) 28 days, XPS spectra with fitting profiles for the sample with 60% glass phase cured at 60 °C for (e) 7 days and (f) 28 days.

decline in BO content from 58% to 49%, accompanied by an increase in NBO from 29% to 35% and a slight increase in silanol groups. Together, the decrease in BO and increase in NBO/Si-OH provided direct evidence for the progressive breakdown of the network. The resulting microstructure became less polymerized and potentially less stable under prolonged high-temperature curing in the absence of sufficient network-forming precursor.

In contrast, for samples with 60% glass phase cured at 90 °C (Fig. 7c and d), the Si 2p_{3/2} binding energy increased marginally from 101.94 eV to 101.98 eV, while the Al 2p_{3/2} binding energy showed a more substantial rise from 73.98 eV to 74.18 eV. This upward shift signified a decrease in local electron density around Si and Al, consistent with their incorporation into a more polymerized, cross-linked network. Correspondingly, the oxygen environment underwent a dramatic reorganization. The BO fraction increased dramatically from 28% to 50%, while the NBO content decreased proportionally from 72% to 50%. This pronounced conversion of NBO to BO underscored the critical role of glass phase in supplying reactive silica, thereby promoting extensive polycondensation to form a continuous, cross-linked gel.

For the same glass-rich sample cured at the lower temperature of 60 °C (Fig. 7e and f), a similar but more gradual trend was observed. Si 2p_{3/2} (101.83 eV to 101.90 eV) and Al 2p_{3/2} (73.91 eV to 73.93 eV) binding energies increased slightly. The BO content increased from 46% to 55%, whereas the NBO content decreased from 42% to 33%. The silanol concentration remained largely unchanged. This evolution suggested that NBOs were steadily consumed to form new bridging bonds, confirming a continuous yet slower polycondensation process that progressively extended the aluminosilicate network.

3.7. Fourier-transform infrared spectra

The FT-IR spectra of samples cured at 90 °C and 60 °C for 7 and 28 days were presented in Fig. 8. All samples exhibited highly similar spectral features, indicating consistent chemical environments and phase compositions across different samples. Characteristic absorption bands were assigned as follows: the broad band centered at 3439 cm⁻¹ was attributed to O-H stretching vibrations influenced by hydrogen bonding of varying strength in water molecules or structural hydroxyl groups. The sharp peak at 1644 cm⁻¹ was assigned to the vibrations of hydroxyl groups M-O-H in surface adsorbed water in geopolymers. A prominent broad absorption envelope in the 900–1100 cm⁻¹ region arose from asymmetric and symmetric stretching vibrations of Si-O-T (T = Si or Al) bonds in the N-A-S-H gels framework. The principal vibration occurred near 992 cm⁻¹. With increasing glassy phase content in the precursors, this main peak shifted systematically to higher

wavenumbers, suggesting an increase in the silicate network polymerization degree and a more cross-linked, stable gel microstructure. In the fingerprint region, several discrete peaks indicated the coexistence of crystalline zeolitic phases. The band at 735 cm⁻¹ was assigned to symmetric stretching of four-membered ring structures in ZK-14 zeolite, a vibration that was also commonly observed in the symmetrical vibration of Si-O-T bonding of AlO₄ and SiO₄ tetrahedra in geopolymer structures. The characteristic peaks observed at 669 cm⁻¹ and 444 cm⁻¹ were attributable to ZK-14 zeolite, corresponding to the single/double six-membered ring structures and O-T-O (T = Si or Al) bending modes in the S4R zeolite structure, respectively. Furthermore, the absorption at 583 cm⁻¹ fell within the typical range for zeolitic framework vibrations and was consistent with the presence of Na-P1 zeolite, likely associated with double rings vibration. A weak but discernible band near 1420 cm⁻¹ was identified as the symmetric stretching mode of C-O bonds in carbonate species. Its presence confirmed slight carbonation of the samples, resulting from the reaction between residual alkali and atmospheric CO₂ [44,46].

3.8. Microstructure

Fig. 9 revealed the microstructures of samples with varying glass phase contents cured at different temperatures. After 7 days at 60 °C, all samples featured typical zeolite crystal morphologies, consisting of agglomerated rectangular prismatic crystals along with numerous elongated rod-like or fibrous forms. These crystals interwove and clustered into a porous network. Within the voids and on the surfaces of these crystal networks, amorphous N-A-S-H gels were clearly present, filling and binding the crystalline networks to form the microstructure of geopolymers.

In contrast, at 90 °C, the microstructure underwent a marked transformation. In the sample with 30% glass phase content (Fig. 9i), unique petal-shaped crystals developed and grew extensively within the matrix. EDS mapping analysis (Fig. 9j) confirmed that these petal-shaped regions were rich in Na, Al, Si, and O. Notably, their growth was accompanied by numerous cracks and pores in the surrounding matrix. This observation directly supported that the excessive growth of zeolite crystals under high-temperature curing may generate localized stress within the matrix, thereby inducing microcracks. This was proposed as a key microstructural mechanism behind the observed late-stage strength regression. The curing temperature had a decisive influence on zeolite morphology.

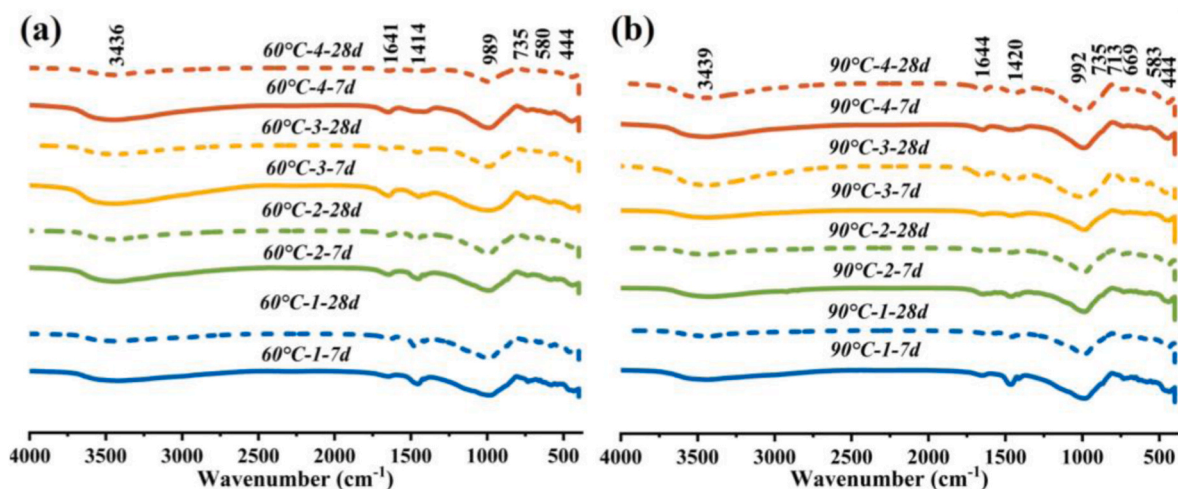


Fig. 8. FT-IR spectra for samples of varying composition under (a) 60 °C and (b) 90 °C curing.

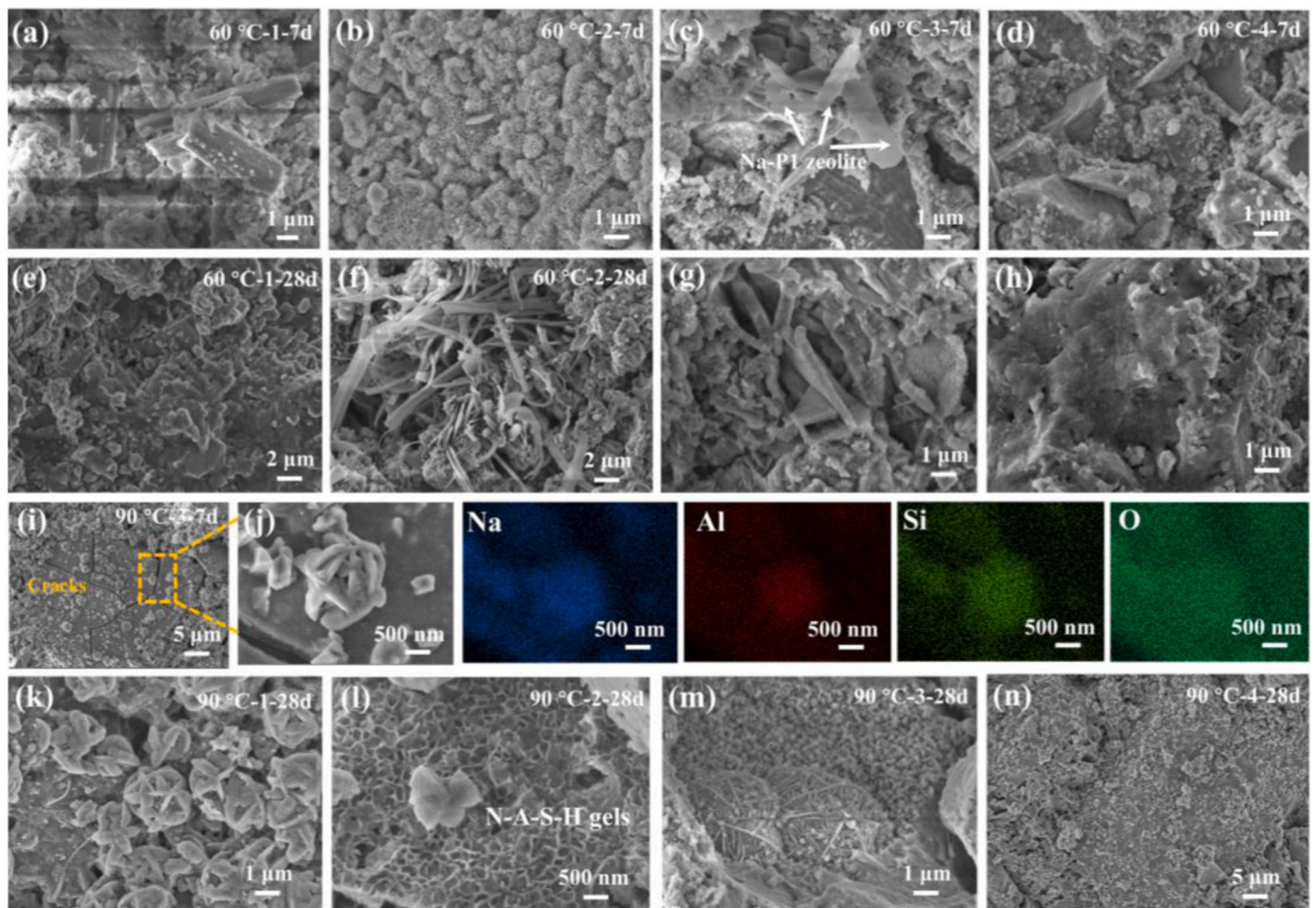


Fig. 9. Morphology and elemental distribution of samples with varying glass phase content under different curing conditions; (a-d) Secondary electron (SE) images of samples with different glass phase contents after curing at 60 °C for 7 days, (e-h) SE images of samples with different glass phase contents after curing at 60 °C for 28 days, (i, j) SE image and corresponding EDS elemental maps (e.g., showing the distributions of Na, Al, Si, and O) of the sample with 30% glass phase after curing at 90 °C for 7 days, (k-n) SE images of samples with different glass phase contents after curing at 90 °C for 28 days.

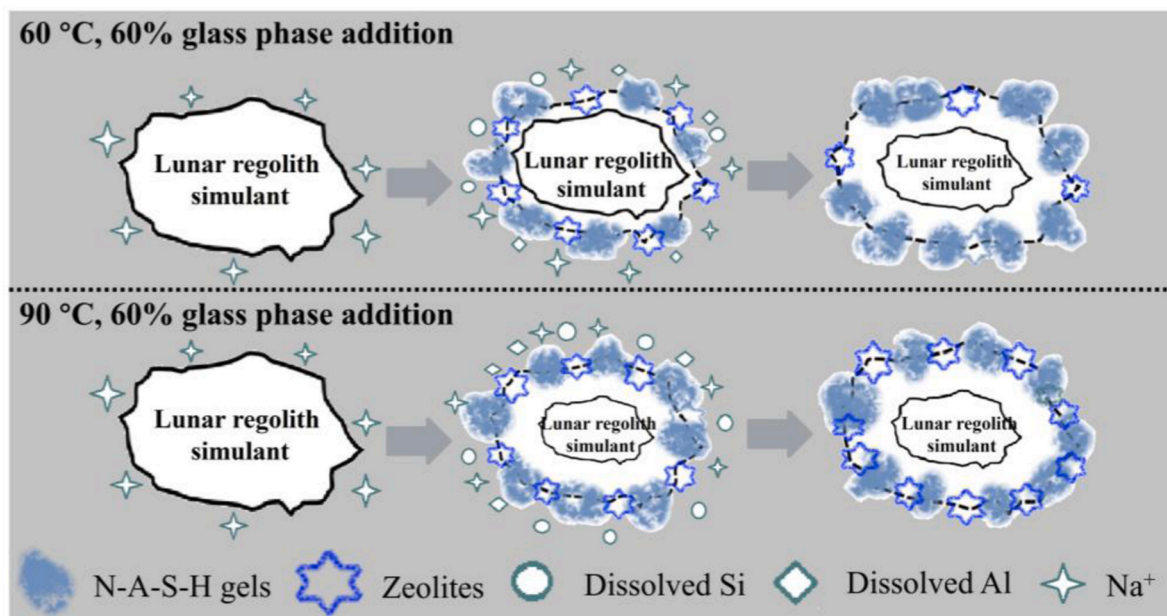


Fig. 10. The schematic comparison of the samples at different temperatures incorporating 60% glass phase in the hydration process.

4. Discussion

4.1. Mechanistic illustration of reaction pathways

Fig. 10 schematically compared the reaction pathways of the sample containing 60% glass phase under the two curing conditions. The reaction could be divided into three stages: initial, middle, and late. At 90 °C, the elevated temperature markedly accelerated the dissolution-polycondensation kinetics during the initial stage, promoting rapid formation and early structural solidification of the N-A-S-H gel. Simultaneously, higher thermal energy promoted the nucleation and initial growth of various zeolitic phases. As the reaction proceeded, continued heating provided sustained driving force for zeolite crystallization. The zeolite crystallization was often accompanied by localized volume changes and internal stresses, potentially inducing micro-cracks and other defects. However, in the sample with high glass phase, the abundant supply of soluble Si and Al supported the extensive formation of a continuous N-A-S-H gel. This network effectively filled pores and bridged micro-cracks, thereby maintaining overall structural stability and enabling mechanical strength to continue increasing in later stages.

Conversely, at 60 °C, the reaction proceeded more moderately. During the middle stage, N-A-S-H gel coexisted with certain zeolitic phases. Over time, these zeolitic species progressively dissolved and reorganized, contributing to the growth of a more stable and continuous gel network. Consequently, the microstructure evolved steadily toward densification without significant crystallization-induced stress, resulting in a monotonic increase in compressive strength throughout the curing period.

4.2. Synergistic mechanism between glass content and curing temperature

This study revealed a clear synergistic relationship between glass content and curing temperature that governed the reaction pathway and final properties of alkali-activated lunar regolith geopolymers. The key insight was that temperature determined the stability and role of zeolites, while glass content controlled the extent of gel formation that ultimately governed microstructural development.

At 60 °C, zeolites initially crystallized as rectangular prismatic or elongated rod-shaped crystals interwoven with the developing N-A-S-H gel. However, under prolonged curing at this temperature, these zeolites exhibited thermodynamic metastability. This was evidenced by the gradual weakening of characteristic diffraction peaks over time. As curing proceeds, the progressive formation of N-A-S-H gel dominated the microstructural evolution. This continued gel production was evidenced by the gradual refinement of pore structure, a steady increase in bridging oxygen (BO) content, and sustained strength development.

At 90 °C, zeolites crystallized as thermodynamically stable phases that persisted throughout curing. However, their impact on performance depended critically on glass content. In low-glass samples, insufficient gel formation allowed disordered, petal-shaped zeolites to disrupt the matrix, leading to microstructural coarsening, a decline in BO content, and late-age strength loss. In high-glass samples, extensive early-age gel formed a highly cross-linked network that enveloped and integrated the stable zeolites into a dense matrix, enabling continued densification and sustained strength gain.

5. Conclusion

The glass phase, as the most reactive amorphous component in lunar regolith, constitutes a critical material for in-situ resource utilization in lunar construction. This study comprehensively elucidated the critical and synergistic effects of glass-phase content and curing temperature on the reaction pathways, microstructural evolution, and ultimate mechanical performance of alkali-activated lunar regolith simulants. A multi-technique microstructural investigation, incorporating compressive strength testing, XRD, TG-DTG, MIP, XPS, and SEM, was employed

to track phase evolution, pore structure development, and morphological changes. This integrated approach revealed clear relationships between processing conditions, microstructure and mechanical properties. The main findings are summarized below.

- 1) At 60 °C, increased glass content promoted continuous strength development, accompanied by gradually reducing zeolitic phases over time. In contrast, at 90 °C, glass phase content became decisive for long-term performance. While all mixes achieved high early strength, only samples with high glass-phase content (SH-4, 60% glass) maintained progressive strengthening over time. Those with low glass content exhibited a slight late-stage strength regression. This decline was attributed to the persistence of zeolitic phases at elevated temperature, highlighting the temperature-dependent stability of these zeolites and their competitive role in microstructural evolution.
- 2) At both curing temperatures, increasing glass phase content effectively reduced total porosity. At 60 °C, porosity continued to optimize with aging, primarily due to the gradual reduction of large pores (>1000 nm) and capillary pores (100-1000 nm), which were harmful to strength. At 90 °C, the 28-day porosity of samples with low glass phase content was higher than at 7 days. Conversely, samples with 60% glass phase content exhibited the opposite trend, showing lower 28-day porosity than at 7 days. This confirmed the critical role of high glass phase content in promoting microstructural densification at elevated temperatures.
- 3) The glass-free sample cured at 90 °C exhibited a significant decrease in Si 2p and Al 2p binding energies, together with a decline in BO content from 58% to 49% and a concurrent rise in NBO content. These changes confirmed the progressive depolymerization of the aluminosilicate network, explaining the observed microstructural coarsening and long-term strength loss. Conversely, the sample containing 60% glass phase cured at 90 °C showed a marked increase in Al 2p binding energy and a substantial rise in BO fraction (from 28% to 50%). This shift reflected a pronounced “NBO-to-BO” conversion, driven by the abundant reactive silica supplied from the glass phase, which facilitated extensive polycondensation and the formation of a highly cross-linked gel network. This chemically stable matrix was the foundation for superior durability and strength. At 60 °C, the evolution of binding energies and BO content (from 46% to 55%) proceeded more gradually, indicating a slower yet steady polycondensation process consistent with the corresponding strength development.
- 4) Microstructural development analysis revealed that samples curing at 60 °C formed rectangular prismatic or elongated rod-shaped zeolite crystals interwoven with N-A-S-H gel, resulting in a relatively ordered microstructure. However, at 90 °C, the crystal morphology shifted to disordered, petal-shaped zeolite that grew and interpenetrated randomly within the matrix. This disorderly growth introduced localized stress concentrations and microcracks, compromising the structural integrity and continuity, and was identified as the key microstructural factor leading to late-stage strength regression.

This study successfully optimized the reaction pathways and microstructure of alkali-activated lunar regolith simulants by synergistically regulating glass phase content and curing temperature, demonstrating the critical role of high glass phase content for suppressing strength degradation at elevated temperature and promoting a densified gel network. Nevertheless, this study has several limitations that should be acknowledged. First, the lunar regolith simulant used in this study, while carefully characterized and compositionally similar to genuine lunar regolith in many aspects, inevitably differs from actual lunar materials in several key features. These include space weathering characteristics, the presence of nanophase iron, complex mineral assemblages, and particle size distribution. Such differences may influence

the reactivity and geopolymerization behavior under lunar conditions. Additionally, only a single source of basaltic simulant was employed, which does not capture the compositional diversity of different lunar regions, such as mare and highland sites. Furthermore, all curing was performed under ambient laboratory conditions without simulating lunar vacuum, thermal cycling, or radiation. Future work should therefore employ lunar regolith simulants that more closely replicate the characteristics of genuine lunar regolith, extend the investigation to a wider range of simulants representing different lunar sites, and conduct experiments under simulated lunar environmental conditions.

CRedit authorship contribution statement

Zhengzheng Liu: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Yukun Qin:** Methodology. **Chuanlin Hu:** Supervision, Funding acquisition, Conceptualization, Writing – original draft, Writing – review & editing. **Huajun Sun:** Funding acquisition. **Xueping Peng:** Supervision. **Fazhou Wang:** Funding acquisition, Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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